OPENS AND QUASI-COMPACT OPENS OF A COFILTERED LIMIT OF TOPOLOGICAL SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The construction of the limit of a diagram of topological spaces over a cofiltered category, as the subset of the product with the induced topology. Given in the author's section on limits of spaces, immediately before the counted statement. Supplies the description of the limit and its projection maps in terms of which both assertions are formulated.

Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered category. Let $i \mapsto X_i$ be a diagram of topological spaces over $\mathcal{I}$. Let $X = \lim X_i$ be the limit with projection maps $f_i : X \to X_i$.*

1. *Any open of X is of the form $\bigcup_{j \in J} f_j^{-1}(U_j)$ for some subset $J \subset I$ and opens $U_j \subset X_j$.*
2. *Any quasi-compact open of X is of the form $f_i^{-1}(U_i)$ for some i and some $U_i \subset X_i$ open.*

Proof. The construction of the limit given above shows that $X \subset \prod X_i$ with the induced topology. A basis for the topology of $\prod X_i$ are the opens $\prod U_i$ where $U_i \subset X_i$ is open and $U_i = X_i$ for almost all i. Say $i_1, \ldots, i_n \in \mathrm{Ob}(\mathcal{I})$ are the objects such that $U_{i_j} \neq X_{i_j}$. Then

$$X \cap \prod U_i = f_{i_1}^{-1}(U_{i_1}) \cap \ldots \cap f_{i_n}^{-1}(U_{i_n})$$

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For a general limit of topological spaces these form a basis for the topology on X. However, if $\mathcal{I}$ is cofiltered as in the statement of the lemma, then we can pick a $j \in \mathrm{Ob}(\mathcal{I})$ and morphisms $j \to i_l$, $l = 1, \ldots, n$. Let

$$U_j = (X_j \to X_{i_1})^{-1}(U_{i_1}) \cap \ldots \cap (X_j \to X_{i_n})^{-1}(U_{i_n})$$

Then it is clear that $X \cap \prod U_i = f_j^{-1}(U_j)$. Thus for any open W of X there is a set A and a map $\alpha : A \to \mathrm{Ob}(\mathcal{I})$ and opens $U_a \subset X_{\alpha(a)}$ such that $W = \bigcup f_{\alpha(a)}^{-1}(U_a)$. Set $J = \mathrm{Im}(\alpha)$ and for $j \in J$ set $U_j = \bigcup_{\alpha(a)=j} U_a$ to see that $W = \bigcup_{j \in J} f_j^{-1}(U_j)$. This proves (1).

To see (2) suppose that $\bigcup_{j \in J} f_j^{-1}(U_j)$ is quasi-compact. Then it is equal to $f_{j_1}^{-1}(U_{j_1}) \cup \ldots \cup f_{j_m}^{-1}(U_{j_m})$ for some $j_1, \ldots, j_m \in J$. Since $\mathcal{I}$ is cofiltered, we can pick a $i \in \mathrm{Ob}(\mathcal{I})$ and morphisms $i \to j_l$, $l = 1, \ldots, m$. Let

$$U_i = (X_i \to X_{j_1})^{-1}(U_{j_1}) \cup \ldots \cup (X_i \to X_{j_m})^{-1}(U_{j_m})$$

Then our open equals $f_i^{-1}(U_i)$ as desired. $\square$